

UKIIVIK, Greenland

WMO: 042530

Lat: 62.578N

Lon: 50.406W

Elev: 22

StdP: 101.06

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.7	-14.9	-22.7	0.5	-16.0	-20.6	0.6	-14.1	23.7	-5.5	22.1	-7.2	6.9	70	0.990

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	11.1	8.3	9.7	7.7	8.7	7.1	8.9	10.2	8.1	9.2	7.5	8.4	4.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	6.8	9.1	7.4	6.4	8.4	6.8	6.2	7.8	27.0	10.3	25.2	9.3	23.8	8.5	13.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.8	15.2	DB	-16.9	15.0	3.2	2.6	-19.2	16.9	-21.1	18.5	-22.9	19.9	-25.2	21.8	(4)
(5)			WB	-17.6	10.8	2.9	1.1	-19.6	11.6	-21.3	12.3	-23.0	12.9	-25.1	13.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-5.3	-5.8	-5.2	-1.6	1.7	4.3	5.8	6.0	4.3	1.5	-1.5	-3.7	
	DBStd	5.43	4.58	5.20	4.96	3.11	2.37	2.13	1.95	1.78	1.70	2.78	3.29	3.92	
	HDD10.0	3627	476	443	471	347	258	172	131	126	171	264	346	425	
	HDD18.3	6667	734	677	729	597	516	421	388	384	421	522	596	683	
	CDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	6.4	7.5	7.8	7.8	7.1	5.9	5.2	4.9	4.8	5.6	6.2	7.1	7.3	
Precipitation	PrecAvg	1067	111	83	75	71	60	58	62	91	109	94	123	130	
	PrecMax	1836	640	212	226	203	134	151	210	199	204	212	445	467	
	PrecMin	518	14	12	4	17	15	6	5	21	24	16	18	46	
	PrecStd	322	111	54	50	43	34	36	46	49	46	52	94	96	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	7.2	6.9	7.4	10.8	12.1	13.1	12.2	12.0	11.4	8.4	6.4	
		MCWB	2.8	3.0	3.7	3.2	6.5	8.0	9.5	9.4	6.9	6.1	5.1	2.5	
	2%	DB	4.0	4.7	4.0	5.0	7.6	9.7	10.6	10.4	8.4	7.5	6.0	3.8	
		MCWB	1.7	1.6	1.9	2.4	5.3	7.7	8.5	8.5	6.3	4.5	3.2	1.3	
	5%	DB	1.9	2.5	2.4	3.4	6.0	8.3	9.4	9.2	7.3	5.9	4.0	2.7	
		MCWB	0.2	0.5	1.0	2.0	4.6	6.7	7.9	7.9	5.8	4.1	2.3	1.0	
	10%	DB	0.4	0.8	1.0	2.3	4.8	7.1	8.4	8.3	6.5	4.7	2.6	1.4	
		MCWB	-0.7	-0.4	0.1	1.3	3.7	6.0	7.3	7.4	5.3	3.4	1.4	0.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.9	3.9	4.2	7.2	9.6	10.2	10.1	8.2	7.1	5.8	3.0
			MCDB	5.2	6.4	5.8	5.4	9.1	11.3	12.1	11.5	9.5	10.2	8.1	4.8
2%		WB	1.9	2.4	2.4	3.0	5.8	7.8	8.9	8.9	7.0	5.2	3.8	2.0	
		MCDB	4.0	4.1	3.6	4.4	7.1	9.2	10.0	9.8	8.0	6.7	5.2	3.5	
5%		WB	0.5	0.8	1.1	2.2	4.7	6.9	8.1	8.1	6.1	4.3	2.4	1.2	
		MCDB	1.6	2.0	2.0	3.3	5.7	8.1	9.1	8.9	7.0	5.4	3.7	2.3	
10%		WB	-0.6	-0.4	0.1	1.4	3.7	6.0	7.4	7.5	5.5	3.6	1.5	0.2	
		MCDB	0.4	0.8	0.9	2.2	4.6	7.0	8.3	8.2	6.2	4.6	2.6	1.2	
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.2	4.0	3.3	2.9	3.2	3.1	2.8	2.6	2.8	3.1	3.6	
		MCDBR	5.4	5.9	5.1	4.9	5.0	5.4	4.8	4.3	4.1	4.6	4.8	4.9	
		MCWBR	4.1	4.1	3.8	3.5	3.1	3.4	3.0	3.0	2.8	2.9	3.7	3.7	
	5% WB	MCDBR	5.3	5.8	4.8	4.5	4.5	5.0	4.3	3.9	3.5	3.9	4.5	4.4	
		MCWBR	4.3	4.6	3.8	3.4	3.1	3.5	3.0	3.0	2.7	2.8	3.9	3.6	
Clear-Sky Solar Irradiance	taub	0.222	0.264	0.288	0.329	0.345	0.347	0.352	0.351	0.315	0.278	0.214	0.360		
	taud	2.026	2.273	2.343	2.274	2.319	2.390	2.396	2.417	2.456	2.356	2.079	2.832		
	Ebn at Noon	434	706	829	855	870	875	859	827	794	678	446	312		
	Edh at Noon	34	58	80	105	110	105	102	92	72	53	30	20		
All-Sky Solar Radiation	RadAvg	0.17	0.72	2.02	3.67	4.93	5.40	4.97	3.60	2.34	1.16	0.31	0.09		
	RadStd	0.02	0.12	0.21	0.45	0.39	0.47	0.50	0.37	0.30	0.13	0.06	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.57	+2.05	N/A	+0.93	N/A	N/A	N/A	N/A	+11	N/A

Nomenclature: See separate page